



www.tfreemantle.com

T.FREEMANTLE LIMITED
13 ATKINSON WAY
FOXHILLS INDUSTRIAL ESTATE
SCUNTHORPE
NORTH LINCOLNSHIRE
UNITED KINGDOM
DN15 8QJ

T: +44 (0) 1724 276 908
F: +44 (0) 1724 276 909
E: sales@tfreemantle.com

SEMI AUTOMATIC END LOAD CARTONER (S07)



OPERATIONS MANUAL

S07 END LOAD CARTONER

USER INSTRUCTION MANUAL

READ THIS MANUAL CAREFULLY BEFORE OPERATION



STANDARD FEATURES

- ROTARY CARTON FEED SYSTEM
- MIMIC BOARD FAULT INDICATOR ON L.E.D DISPLAY
- DIGITAL PRODUCTION COUNTER
- VARIABLE SPEED MACHINE CONTROL FROM CONTROL POD
- OVERLOAD PROTECTION ON MACHINE MAIN DRIVE
- DIGITAL READOUTS TO ASSIST CHANGE SIZE
- DISCHARGE DRIVE BELTS
- LOW LEVEL MAGAZINE CARTON DETECTION
- AUTOMATIC CUT-OFF SWITCH, IF CARTON NOT FULLY LOADED
- JOG AND RESET BUTTON ON WANDERING LEAD
- GLUE LOW TEMPERATURE INTERLOCK (WHEN FITTED)
- LOW AIR PRESSURE SWITCH

CONTENTS

Section 1: General Information

Page

A	Introduction&Safety	
	1. Preface	7
	2. Safety Instructions	
	a) General Information	8
	b) Operation	9
	c) Maintenance	9/10
	d) Cleaning	10
	e) Liability	11
B	Technical Data	
	1. Dimensions	12
	2. Output	12
	3. Format Range	12
	4. Machine Data&Requirements	13
	5. Sound Emission Data	14
C	Description of Machine	
	1. General Description	
	a) Machine Construction	15
	b) Machine Specification	15
	1. Identification	16
	2. Operating Panel (Pod)	16
	3. Machine Functions	17
	a) Product Infeed	18
	b) Product Loading	18
	c) Carton Closure	19
	i) Gluing	19
	ii) Tucking	20
	d) Carton Discharge	20
D	Installation&Connection	21

Section 2: Operation

A Initial Operation

- 1. Preparing Machine for Operation**
 - a) Main Control Panel** 22
 - b) Pre Production Check list** 22/24
- 1. Quick Start Instructions** 25
- 2. Operator Input** 26
- 3. Housekeeping**
 - a) Routine Cleaning** 26
 - b) Cartons** 27
 - c) Adhesive** 27

B Trouble Shooting

- 1. Machine Will Not Run** 28
- 2. Carton Erection** 28/29
- 3. Vacuum Failure** 29
- 4. Carton Closure Issues** 29/30
- 5. Dust Flap Closure** 30
- 6. Clutch Overload** 31

Section 3: Maintenance and Setting

A Maintenance Schedule

1. Lubrication Schedule	32
2. Maintenance Schedule	33/34
3. Air pressure Regulator	34
4. Vacuum Pump	35
5. Electrical System	36

B Size Change

Note	37
1. Cartoner Change	37/39
2. Rotary Cartoner Feed Change	40/42
3. Glue System Change	43
4. Tuck System Change	44

Section 4: Appendix

A Hot Glue Melt Applicator

- | | |
|---------------------------------|-------|
| 1. Hazards | 45 |
| 2. Essential Safety Precautions | 45/46 |
| 3. Emergency Actions | 46 |

SECTION 1

A

INTRODUCTION & SAFETY

1.A.1 Preface

This instruction manual is directed to personnel who operate and maintain this machine. This manual should not be regarded as a substitute for a thorough instruction in the installation and operation of the machine.

We recommend that your operating personnel participate in the instruction and training programs provided by T. Freemantle's highly skilled service personnel. This service is normally available during installation and commissioning.

This manual is meant to help you during your daily work on the machine and during maintenance, which must be carried out regularly.

All personnel involved in the installation, set-up, operation, maintenance and/or repair of this machine must thoroughly read and fully understand this instruction manual.

If you should have any questions, please contact T. Freemantle Ltd. The manual is to be stored in a place near to the machine and must be easily accessible for the operator.

Please operate your machine in accordance with the instructions given in this manual. This will guarantee trouble-free operation and a high availability of your machine.

1.A.2 Safety

a) General Information

This machine has been designed and constructed to meet a high technical standard and is therefore safe and reliable in operation.

- Nevertheless the machine can become dangerous
- If operated improperly by personnel not trained and instructed thoroughly.
- If the instructions given in this operation and maintenance manual are not strictly observed.

We specifically point out that you should only

- Operate
- Service
- Maintain
- Supervise

The machine if you have been specially trained and instructed to do so.

The machine must only be operated with all safety devices in their proper location and in their operating order. A Safety Interlock System is fitted to all opening guards, operating at 24 V, inhibiting machine operation if guards are opened.

No modifications must be made to the machine without prior permission of T. Freemantle Ltd.

Note the following when installing the machine:

Hazardous positions between the machine and existing equipment have to be provided with any necessary protection by the customer operating the machine.

Only use products agreed upon with T. Freemantle Ltd. No liability will be accepted for damage arising from improper use and nonobservance of the instructions given in this manual. The user must assume the responsibility.

Hot Melt Glue Applicator Guns (when fitted) are situated within the guarding of the machine, and are electrically heated. The operating temperatures are factory set according to required glue specification.

CAUTION!

*IT IS ADVISED THAT ALL USERS READ THE SAFETY INFORMATION CONTAINED
IN **SECTION 4A** REGARDING HOT MELT ADHESIVE*

WARNING!

DO NOT TOUCH ANY PART OF THE HOT MELT UNIT DURING OPERATION

Note! Care should be taken not to contaminate the glue tanks as this will cause blockages of the glue jet nozzles.

b) **Operation**

Persons must only operate the machine

- Who have been properly trained and instructed in their use.
- Who will perform their duty in reliable manner.

It is incumbent upon the company operating the machine to ensure that the instructions given in this manual are strictly followed. Where necessary the company is to issue supplementary instructions.

c) **Maintenance**

Do not perform any work on the machine while it is in operation. If this should be necessary, however, for example during

- Installation
- Maintenance
- Elimination of faults

Operate the machine or parts of the machine utilising the Jog facility.

Set up a precise working plan if several persons work on the machine at the same time.

Appoint a person who will be responsible

- For coordinating the work to be done
- That safety instructions are strictly followed

The company operating the machine is responsible for providing an efficient safety program for the Superior machine.

The machine must be operated appropriate to its intended use.

The operating procedures applied must be appropriate to the technical possibilities of the machine. Safety and protection devices must not be changed or rendered ineffective.

Check the safety devices on the machine

- At least once a week
- After rebuilding the machine
- After major maintenance or repair work
- After replacing or repairing safety devices

d) Cleaning

The machine is designed to be safe to construct, install, operate and maintain, consistent with current legislation. Consideration has been given to the ergonomics of machinery use, i.e. to achieve the most effective safe performance of the machinery and its operator.

Included in this machine manual are instructions for safe starting up, installation, adjustment, maintenance, training and hints for safe and efficient day-to-day operator input.

T. FREEMANTLE LTD have ensured, so far as is reasonably practicable, that the machinery is so designed and constructed that it will be safe and without risks to health and safety at all times when it is being set, used, cleaned or maintained.

All guarding (fixed or otherwise) should not be removed before or during operation of the machinery. All accessible parts of the machine are guarded (sliding doors or hinged guards) and are electronically safety interlocked. The safety interlock circuit must not be bypassed or overridden.

ROUTINE CLEANING PROCEDURE

Daily	Thoroughly clean the machine to remove any build up of dust, glue and product
Conveyor Chains, Flights etc.	Use a soft brush and clean to remove dust, carton fragments etc.
Infeed Conveyor	Wipe over loading table to remove any product and debris
Machine Guards	Wipe over with stainless steel cleaner and soft cloth
Glue Units	Wipe off excess glue from guides and wheel scrapers etc. Hint! Spraying areas around glue guns with silicone releasing agent before running will aid removal of any excess glue deposits.

e) Liability

You are not allowed to make any changes to the machine without the permission of T. Freemantle Ltd. This includes

- Rebuilding machines, mounting and dismounting parts supplied by other manufacturers.
- Rendering ineffective or removing safety devices.

Any unauthorized modification made on the machine will invalidate the manufacturers liability.

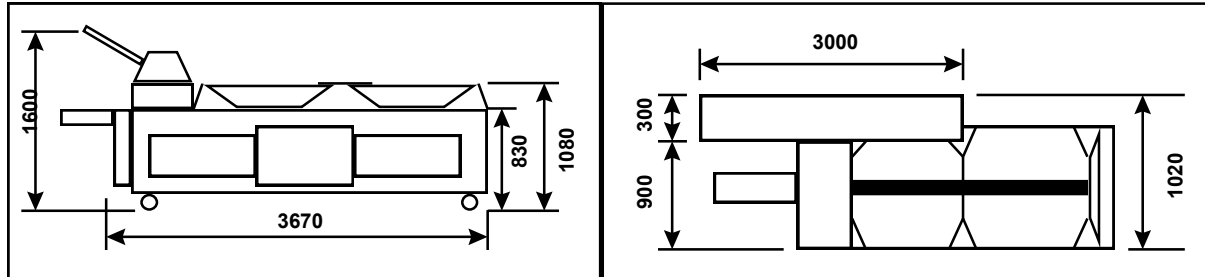
Failure to observe the safety regulations may:

- **Cause danger to the health of the user or to a third party**
- **Cause damage to the equipment or other equipment**
- **Impair the machine's efficiency**

SECTION 1

B TECHNICAL DATA

1.B.1 Dimensions



1.B.2 Output

The machine is fully variable from 25 to 300 cartons per minute, subject to machine flight pitch and size and quality of cartons.

1.B.3 Format Range

The cartoner has a size range as follows, subject to machine pitch and certain restrictions on combinations of sizes.

For Maximum sizes, economy flaps must be used.

6" Pitch

Between Flaps-100/400mm- min/max (3 chain machine)

Between Flaps-125/400mm- min/max (4 chain machine)

Width 50/100mm- min/max

Height 20/100mm- min/max

9" Pitch

Between Flaps-100/400mm- min/max (3 chain machine)

Between Flaps-125/400mm- min/max (4 chain machine)

Width 50/150mm- min/max

Height 20/100mm- min/max

12" Pitch

Between Flaps-100/400mm- min/max (3 chain machine)

Between Flaps-125/400mm- min/max (4 chain machine)

Width 50/250mm- min/max

Height 20/100mm- min/max

15" Pitch

Between Flaps-100/400mm- min/max (3 chain machine)

Between Flaps-125/400mm- min/max (4 chain machine)

Width 50/300mm- min/max

Height 20/100mm- min/max

1.B.4 Machine Data&Requirements

Vacuum Pump Pressure	20 Hg
Vacuum Pump	0.37 kW (Oil less) Ref : VT3.10 B0020
Compressed Air	80 psi / 5.5 bar
Low Air Pressure Switch	40 psi / 2.75 bar
Electrical Power Supply	240 Single - Neutral and Earth, 50 Hz *
AC Inverter Variable Speed	1.5 kW, 2 HP
Infeed Belt Motor (If Fitted)	0.37 kW (0.5 HP) AC Motor. Variable Speed gearbox.
Mag Loader Motor (If Fitted)	0.11 kW (0.125 HP) AC Drum Motor. Inverter control variable speed.
Auto Lube System (If Fitted)	Bielomatik Single Line 'ONE SHOT' Lubrication System (Single Plunger Manually Operated Pump)
Carton Closure	By means of FREEMANTLE DESIGN TUCKING SYSTEM or NORDSON HOT MELT GLUE JET APPLICATOR UNITS
Total Electrical Load	6 kW - Standard Machine

1.B.5 Sound Emission Data

The following illustration shows the decibel noise emission reading from the machine while the machine is in normal continuous production running.

The areas of probable operator input are highlighted on the drawing below, as the machine is a semi-automatic unit there is one defined workstation.

The readings are taken from a point 1 metre from the surface of the machine and 1.6 metres from the floor and are measured in decibels (dB).

(See Fig. 01)

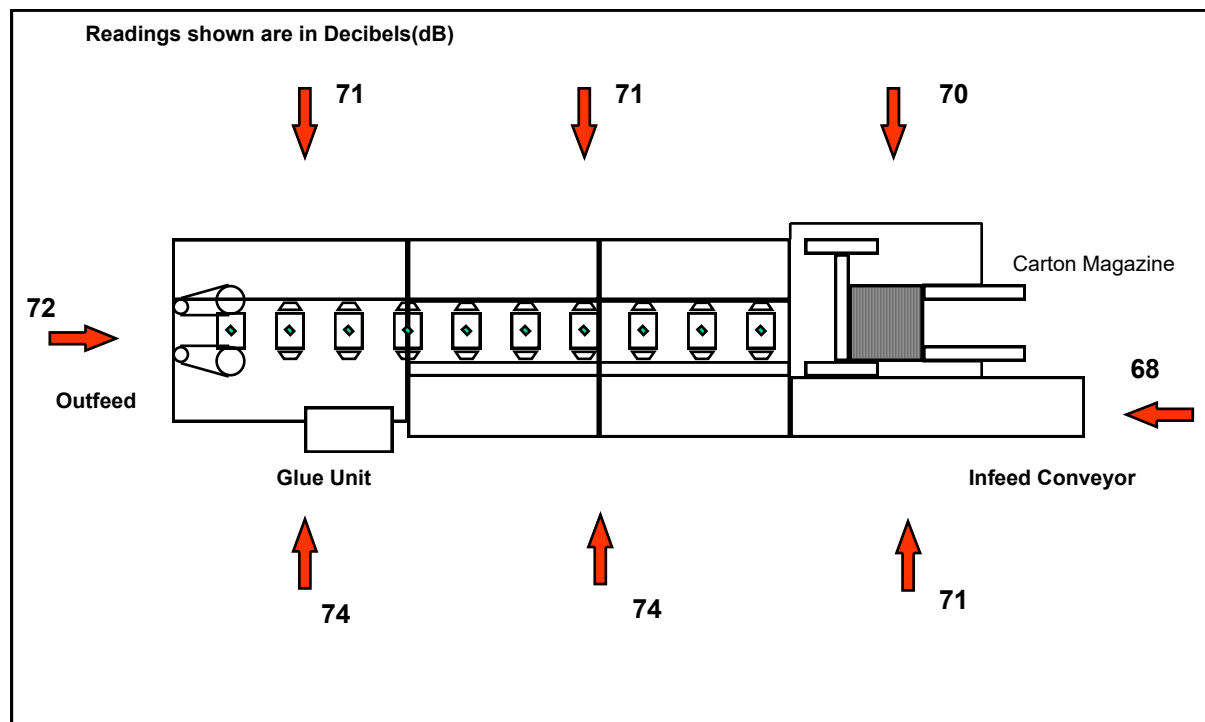


Fig.1

SECTION 1

C DESCRIPTION OF MACHINE

1.C.1 General Description

a) Machine Construction

Construction	Rigid Steel one-piece frame with Stainless Steel and Aluminium Construction. Comprising of Stainless Steel or corrosion resistant coated Shafts, Sprockets, Transmission Drives and Chains as standard unless otherwise stated.
Shafts	Sealed for life Ballraces on all main shafts.
Mobility	Fully mobile on Lockable Castors (Optional Adjustable Jacking Feet).
Guarding	Brush Finish Stainless Steel Guarding, with Polycarbonate / Acrylic viewing stations.

b) Machine Specification

Product Conveyor (Standard Machine)	Bucket or Static Bed product conveyor. 2.0mtr x 300mm Standard machine.
Machine Speed Control	Variable- from operators control pod. Base speed set from main control panel.
Safety Interlock	Full Safety Interlock Guard System- 24V.
Main Drive Protection	Mechanical, Clutch and Electrical Overload Protection on Machine Main Drive.
Discharge Drive Belts	Accelerate cartons efficiently away from path of moving flights.
Low Level Magazine Carton Detection	Linked to warning beacon on Pod.
Automatic Cut-off Switch	Operates if product is not fully loaded.
Low Air Pressure Switch	If air pressure falls below 40 psi, machine will cease to operate.
Jog Facility (Inch)	For fine-tuning of machine. Operates via a wandering lead.
Adjustable Control Panel	Assembly on remote stand, with built in Mimic Board on LED Display.
Change Size Assistance	Digital Readouts to Assist Change Size.
Speed / Product Counter	Digital Readout on operator control pod. (See section 1.C.3)
Rotary Carton Feed System	Twin or Triple Head Rotary Carton Erector for positive erection and placing of carton.
Mimic Board Fault Indicator	Mimic board display to indicate faults machine faults i.e.. Guard Open / Low Air Pressure.

1.C.2 Machine Identification



Fig. 2

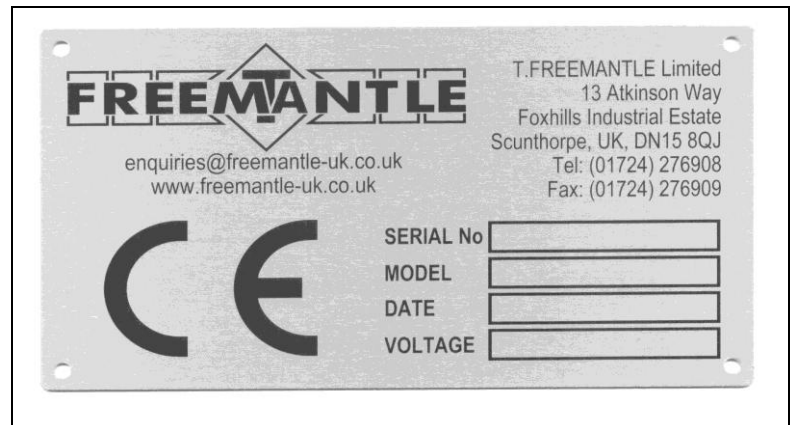


Fig. 3

The CE identification plate is located on the bottom left face of the Main Electrical Panel (Fig. 2/3). Always quote the information on this plate when making technical enquiries.

1.C.3 Operating Panel (Pod)

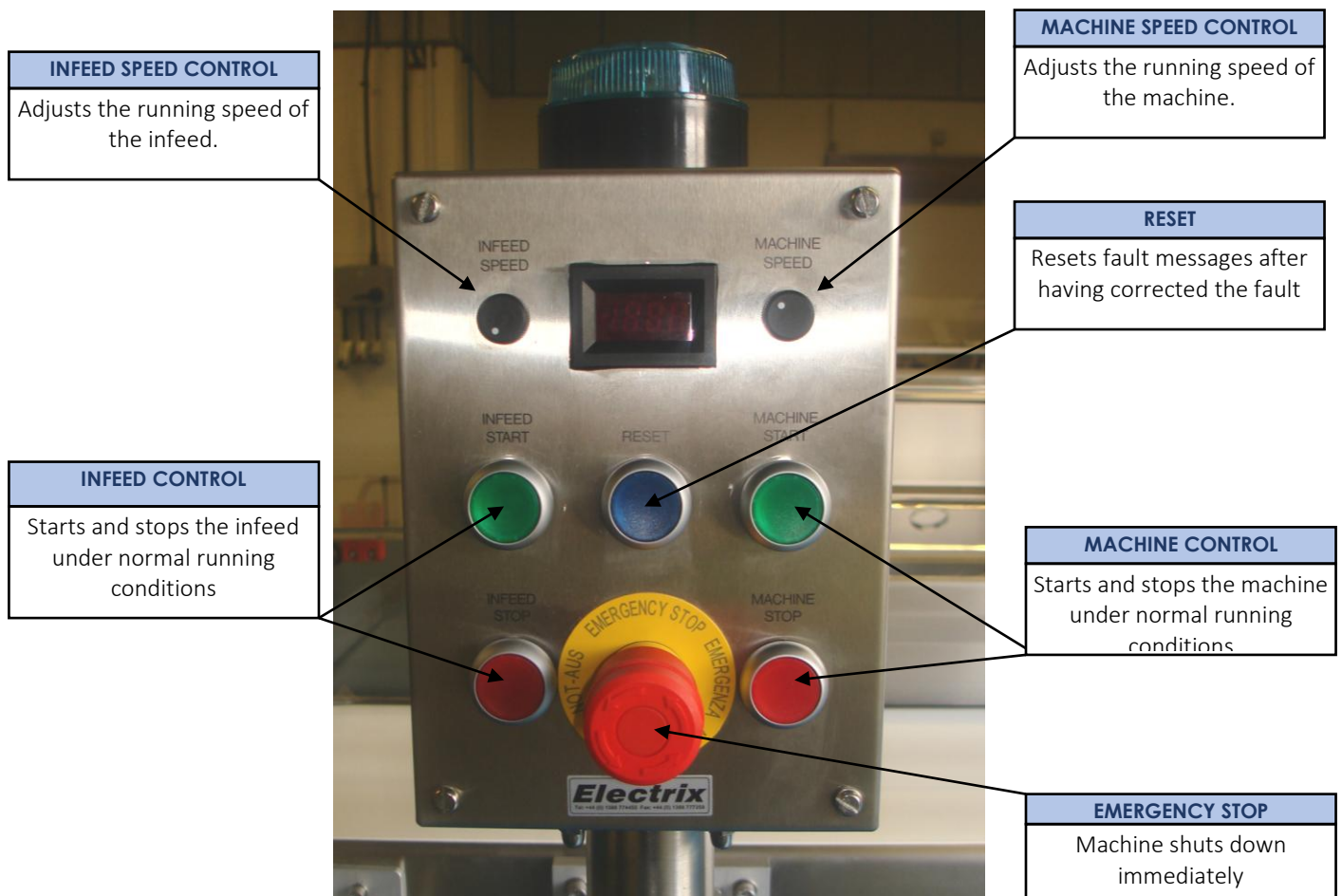


Fig. 4

1.C.4 Machine Functions

Carton blanks are drawn from the angled magazine by a standard twin or triple arm Rotary Carton Feeder and are automatically erected into continuously moving flighted chains. The flighted chains feed the erected cartons towards the operator or loading conveyor. As the cartons are moving, both top and bottom flaps of the load side are automatically spread up and down allowing full access for loading. On the adjustable side of the machine, dust flaps are tucked in and the bottom flap is ploughed up so as to avoid the product being pushed through the carton.

Product is then delivered to the erected carton on the product infeed conveyor and is manually loaded into pre-opened cartons. After loading, cartons are automatically closed and glued or tucked, then discharged by driven side belts (Fig. 5).

Semi Automatic Cartoner

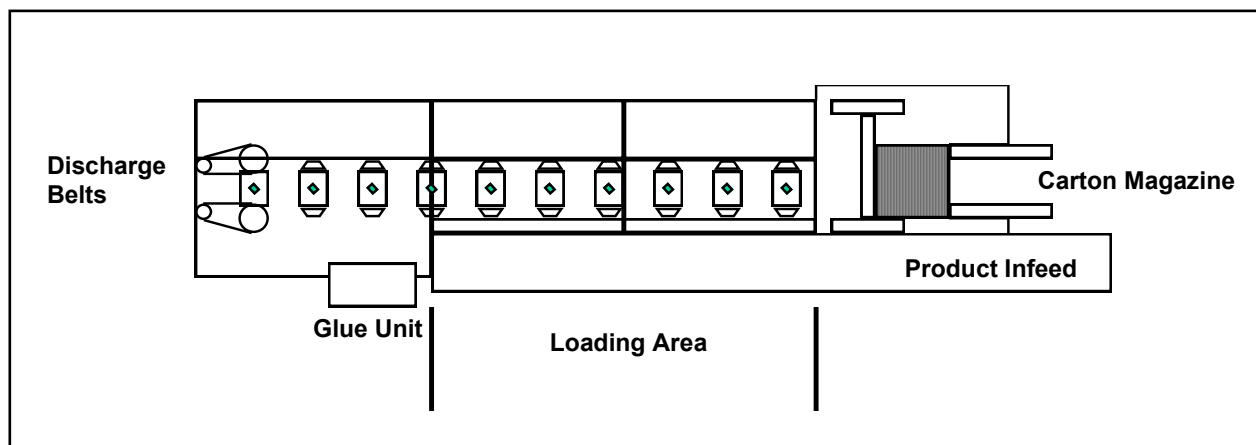


Fig. 5

1.C.4 a) Product Infeed

Product is loaded into the infeed conveyor, which carries the product. The operator inserts the product into the pre-opened carton. (Fig. 6)



Fig. 6

1.C.4 b) Product Loading

The operator pushes the product from the infeed conveyor across the loading table and into the pre-opened carton. The loading table is designed to bridge the product between the product infeed conveyor and the carton mouth and also to condition the product for loading into the carton. It ensures smooth transfer of the product between the conveyor and the carton. (Fig. 7)

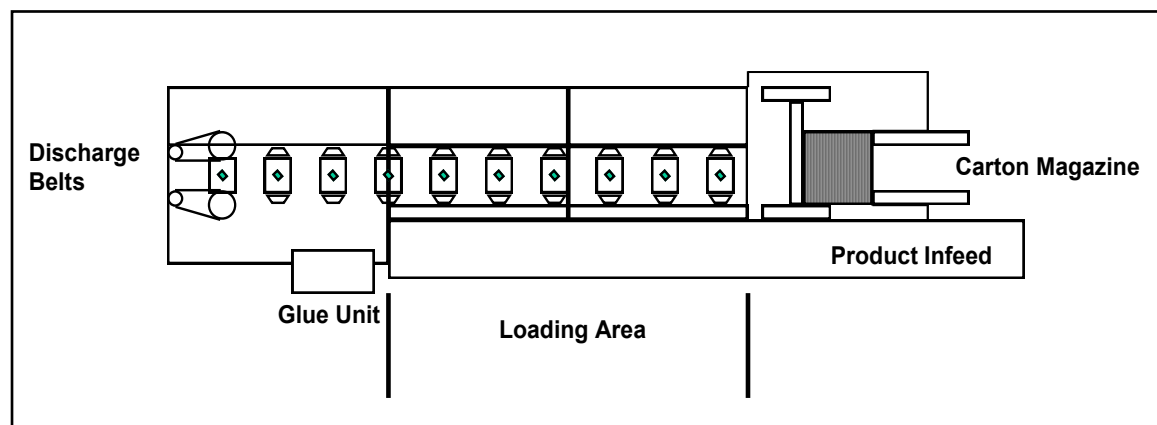


Fig. 7

1.C.4 c) Carton Closure

i) Gluing

The carton enters the gluing station with both bottom flaps folded and top flaps in aeroplane format. (Fig. 8)

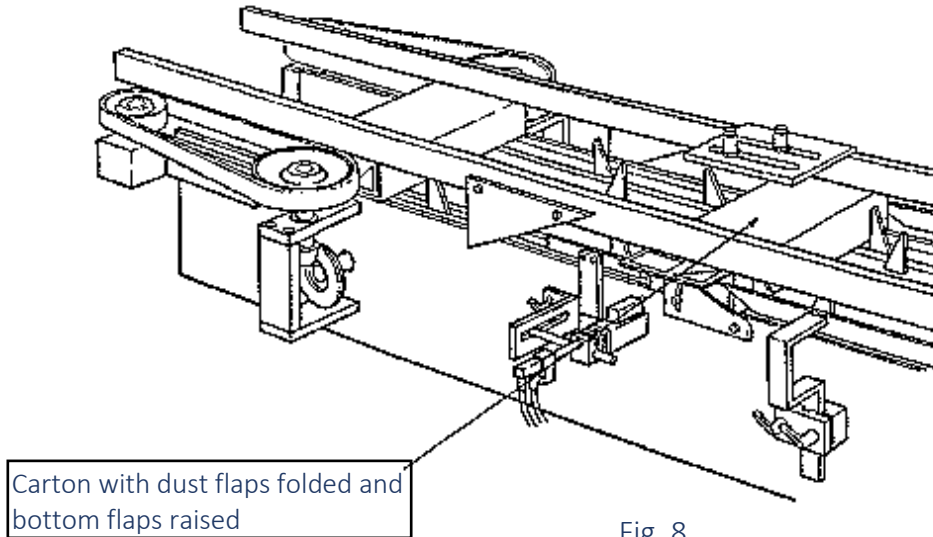


Fig. 8

Glue Line

The length and position of the glue line is controlled by two Photo Electric Cells (P.E.C.s) The 1st sensor detects the leading edge of the carton. 2nd sensor detects the leading edge of the carton and gluing starts. 1st sensor detects the trailing edge and gluing stops. (Fig. 9)

The glue guns should be applying glue to the minor flap so as when the major flap is ploughed down the glue line is approx. 6mm from the edge, when the flap is ploughed down the glue line is compressed to form a good seal. The glue should give a good enough seal so that when the carton is opened the seal should tear the fibres of the carton board.

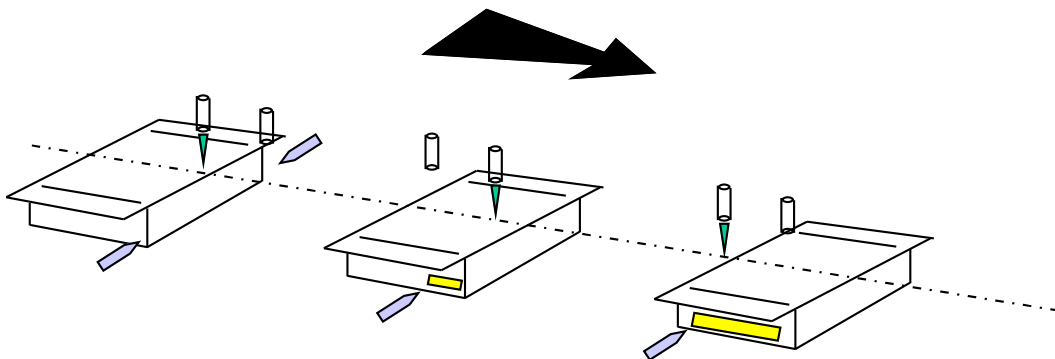


Fig. 9

WARNING!

THE GLUE UNIT WILL BE HOT DO NOT TOUCH WITHOUT ADEQUATE PROTECTIVE CLOTHING

ii) Tucking

Once the product has been loaded and the rotary tuckers close the dust flaps the carton enters the tucking system. This series of guides pre-folds, tucks and finally locks the major flap on the carton (Fig. 10)

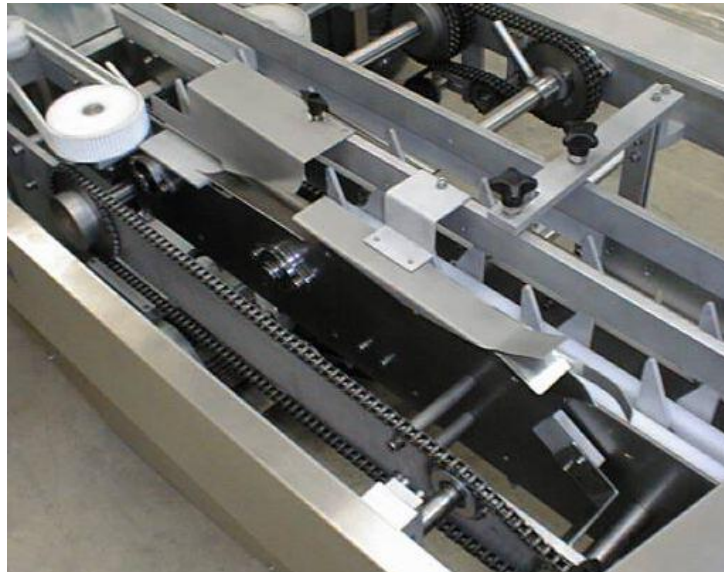


Fig. 10

1.C.4 d) Carton Discharge

After either glue has been applied and the top flaps are automatically ploughed down or the flaps are tucked, the carton is then accelerated out of the machine via the discharge belts.

(Fig. 11)



Fig. 11

SECTION 1

D INSTALLATION & CONNECTION

Transportation

The machine has been designed and manufactured around a rigid steel one-piece frame to give the machines excellent stability and maximum strength. For transportation purposes the machines are generally shipped in one piece. In special circumstances the machines may be shipped with guarding and control pod / panels removed to avoid damage. In the event of the machine needing any components to be removed for shipment then machinery is supplied with information outlining any re-assembly instruction that may be needed.

Installation

This machine has a guarantee of 12 month from date of installation; this guarantee is only applicable if a qualified T. Freemantle engineer installs the machinery.



Allocation should be made for connection to the correct power and compressed air services. (See Machine Data Section 1.B.4)

SECTION 2

A INITIAL OPERATION

2.A.1 Preparing Machine for Operation

a) MAIN CONTROL PANEL

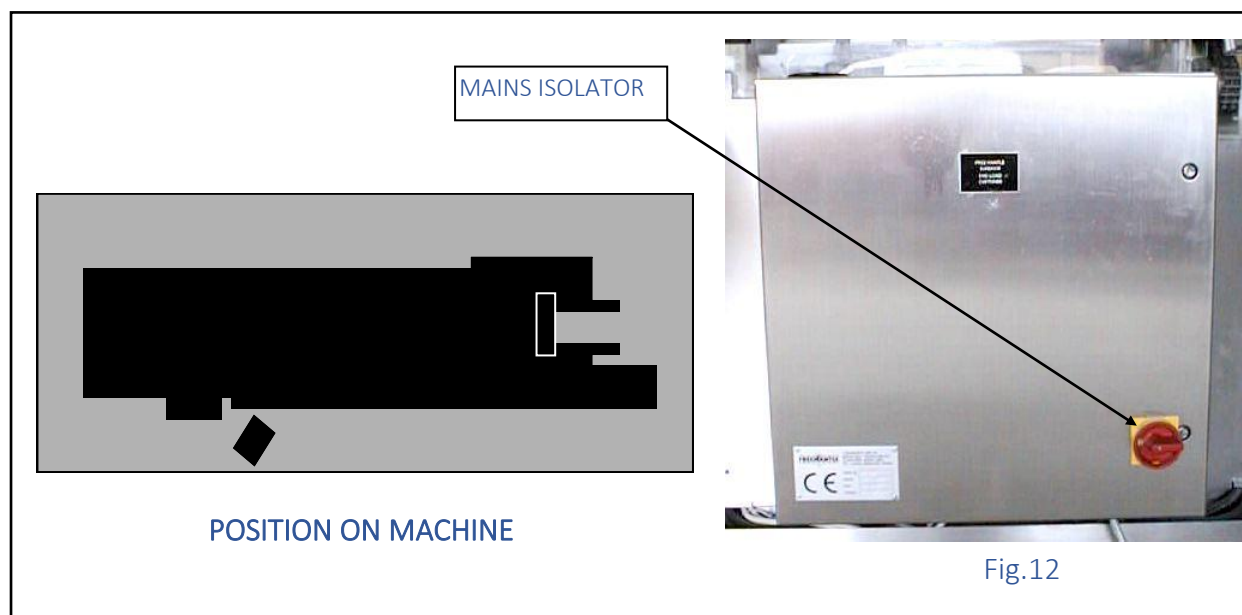
Standard Position

Situated at the Rotary end of the machine.

b) PRE-PRODUCTION CHECK LIST

Electrical Power

Turn Mains Isolator on Control Cabinet to 'ON' position (Fig. 12)



Glue Unit (when fitted)

Machine will not run if the glue is not at the correct operating temperature. When the glue has reached the correct temperature the LED on the control pod will extinguish. We advise that the glue unit is switched on at least 45 min's prior to production to allow the glue to reach its correct operating temperature.

Flight Conveyor

Check that it is empty.

Infeed Conveyor

Check that it is empty.

Hot Melt Glue Tank (when fitted)

Check that it is topped up.

Note! The machine will not run if the glue is not at the correct operating temperature. If the glue level is allowed to drop too low then the introduction of too much fresh (cold) adhesive may lower the glue temperature and stop the machine. Add fresh adhesive little and often.

b) PRE-PRODUCTION CHECK LIST CONTINUED

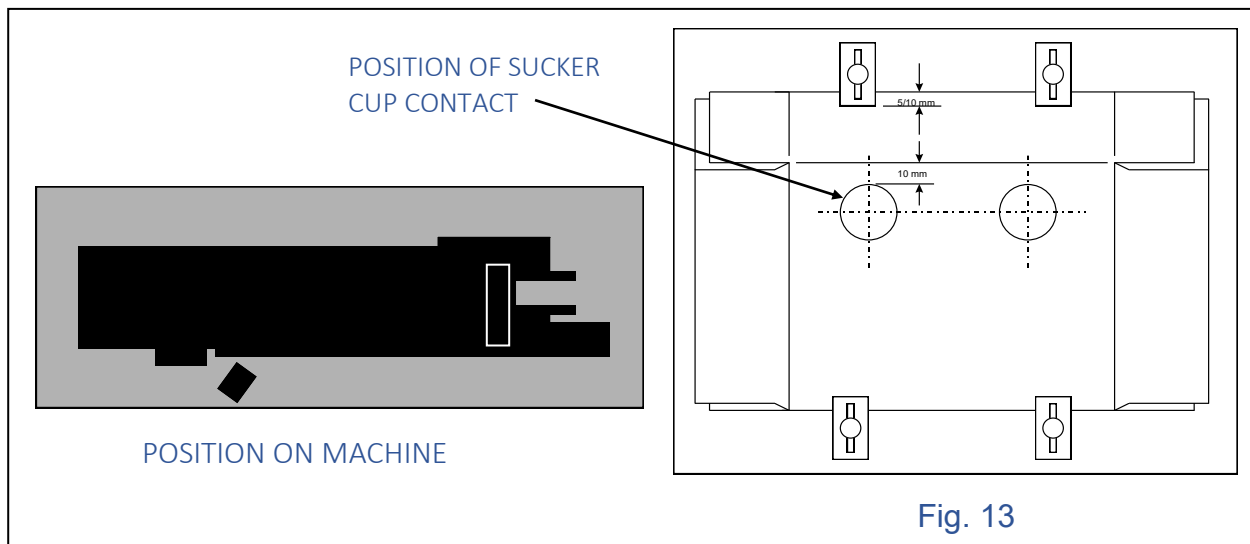
WARNING!

The Glue Tanks must not be contaminated as this will cause blockages of the glue guns

Spraying areas around glue guns with silicone releasing agent before running will aid removal of any excess glue deposits.

Carton Magazine

Check that it is loaded and that the cartons are placed in correctly. (Fig. 13)



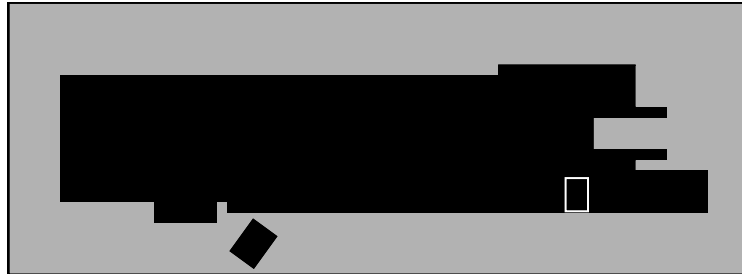
Emergency Stop Buttons

Check that these are released (Fig 14).



b) PRE-PRODUCTION CHECK LIST CONTINUED

Machine Air Supply	Check that the Air Supply is set to the correct level on the Air Regulator Gauge. (Fig. 15)
---------------------------	---



POSITION ON MACHINE



Fig. 15

Guarding	Check that these are closed.
Vacuum Pump	The vacuum pump power's on and off automatically from the footswitch
Reset Safety Switch	Ensure the Reset Push Button lamp lights when pressed.
Start Machine	Press the 'Green Run' Push Button, machine speed can be adjusted by operation of the 'Speed Control Potentiometer.

Jog Remote Lead	Check that it is unplugged
Reset Safety Switch	Ensure all Indicator LED's on the pod are extinguished. Press the 'Illuminated Reset' Push Button and check that the lamp extinguishes.
Start Machine	Press the 'Green Run' Push Button, machine speed can be adjusted by operation of the 'Speed Control Potentiometer.
Product Counter	The counter module fitted to the operators control has the facility to count the amount of product produced. This is measured by the cam switch in conjunction with a photocell mounted by the glue station photocells. Operation of the 'RST' button resets the 'Product Count' to zero.

2.A.2

Quick Start Instructions

OPERATING PROCEDURE	
1	Turn on the Main Isolator on the front of the Main Control Panel.
2	Check that the compressed air is switched on and at the correct temperature. NOTE: - If the machine is a tucking model go to Step 5
3	Ensure that the Glue Tanks are topped up with glue.
4	When the glue has reached its operating temperature the LED's on the control pod will extinguish.
5	Start the Vacuum Pump by pressing the start button on the Pod. NOTE! THE MACHINE <u>WILL</u> RUN WITHOUT THE ABOVE BEING STARTED
6	Close all guards and reset all emergency stop buttons.
7	Check that the carton magazine is full.
8	Check that the 'Remote Lead' is NOT plugged into the main control panel.
9	Press the 'Reset' button on the Pod and observe that its lamp and all LED's extinguish.
10	RUN- Press the "Machine Start". The machine now ramps up to its preset speed. Turning the "Speed Control" Pot on the Pod adjusts the machine speed accordingly.
11	JOG- A JOG facility is available to assist in faultfinding and machine set-up. This is selected by plugging the Remote lead into its socket on the side of the Main Control Panel. The jog pre-set speed is set via the main motor inverter AV1 in the Main Control Panel. Plugging the Remote Lead into its socket on the Main Control Panel allows the jog control on the remote lead only. Also the 'CLUTCH TRIP' LED flashes to indicate that the remote lead is active.

2.A.3 Operator Input

Carton magazine	Operators will need to refill the carton magazine periodically during production.
Product infeed	Loading of the product into the carton is normally undertaken manually to hand feed the product from infeed and into the carton.
Glue unit (when fitted)	During normal production running the glue units will periodically require refilling.
Product discharge	Product is normally taken from the machine automatically and moved on through metal detectors, check weighers etc. Operator input may be required in the event of a product back up.

2.A.4 Housekeeping

Routine Cleaning Procedure

General	Thoroughly clean the machine daily to remove any build up of dust, glue and product.
Conveyor Chains, Flights etc.	Use a soft brush and clean to remove dust, carton fragments etc.
Infeed Conveyor	Wipe over infeed conveyor to remove any product and debris.
Machine Guards	Wipe over with stainless steel cleaner and soft cloth.
Glue Units	Wipe off excess glue from guides and glue guns etc. Hint! Spraying areas around glue guns with silicone releasing agent before running will aid removal of any excess glue deposits.
Out-Feed Belts	Remove any Product or glue deposits.
Carton Guides	Remove any glue deposits or debris from the carton guides and flap ploughs.

Cartons

Carton manufacturers advise that cartons have a shelf life of around three months, providing they are stored away from moisture at temperatures of around 10 / 20 degrees C. If they are stored at varying temperatures or are allowed to become damp, then the shelf life will be reduced considerably, affecting both glue and all creases and therefore seriously reducing the production efficiency of the machine.

When the cartons are stored in cases attention must be given to the number of cases on a stack. If too many cases are stacked together then this may cause the bottom cases to be crushed.

Bent, twisted or damp cartons will cause problems during machine operation.

Adhesive

When any adhesive is used in a factory environment, contamination of that adhesive, both in its raw cold state and in its molten state, it is almost inevitable unless strict housekeeping rules apply. Dirt in the gluing system will give rise to blockage of the glue nozzles, resulting in incorrect glue application.

To avoid contamination keep the glue in its original container completely covered. When filling the glue tanks do not use cartons or such as scoops for transporting the adhesive from the container to the tank as dust or fragments of cardboard could fall into the tank. Dropped adhesive should be swept up and thrown away, do not return it to the container as this will inevitably contain dust and other debris.

Different grades of adhesive should never be mixed and strict control should be maintained on which adhesive is to be used on any specific line or carton.

WARNING!

**HOT MELT ADHESIVE WILL CAUSE SEVERE BURNS DO NOT ATTEMPT TO
CARRY OUT ANY WORK ON OR AROUND THE HOT MELT UNIT WITHOUT
ADEQUATE PROTECTIVE CLOTHING**

SECTION 2

B

TROUBLESHOOTING

2.B.1 Machine Will Not Run

Incorrect Glue Temperature	Check that the glue unit is up to correct operating temp. Check Hot Melt Applicator Unit. (Refer to Manual)
Power Failure	Check that the Emergency Stop buttons are released. Check the mains isolator is 'ON'. Check circuit breakers and main fuses.
Insufficient Air Pressure	Check Air Pressure Regulator Gauge.
Motor Overload	Reset thermal overload on the Inverter.
Safety Interlock Circuit	Check all guards are closed.
Clutch Disengaged	1. Remove product-causing jam. 2. Plug remote 'Jog' lead into the socket on the right hand side of the electrical cabinet. 3. Ensure all guards are closed and the emergency stop buttons are reset. 4. Operate remote 'Jog' button to inch the machine to reset the clutch. 5. When the clutch resets remove the remote lead and continue with normal operation.

2.B.2 Cartons Erection

Rotary Feeder Incorrectly Set	Check and adjust the settings if necessary.
Sucker cups Incorrectly Set	Check and adjust the sucker cups.
Cartons too Tight in Magazine	Check and adjust magazine sides and ledges.
Loss of Vacuum	Check that the sucker cups are not damaged and that there are no leaks in the vacuum system.
Carton Quality	Check if poor, bent, damp, warped or badly glued and contact the supplier.
Cartons not Erecting on Flights	Check timing of Rotary Carton Feeder to Flight Chain (Fig. 16) Check vacuum valve timing is correct (Fig. 17)

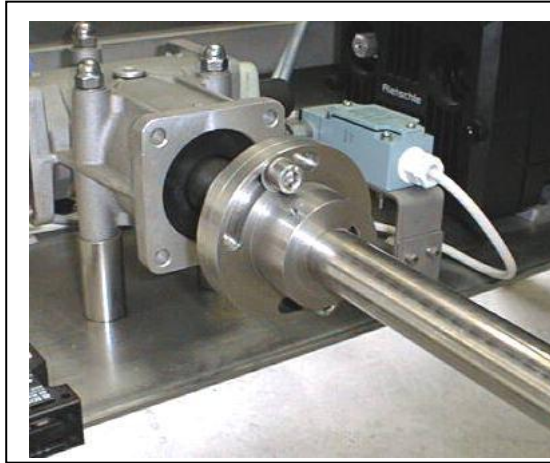


Fig.16

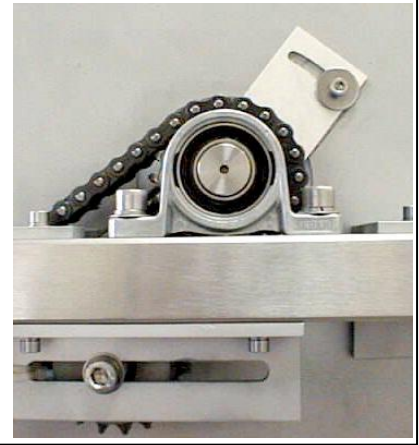


Fig. 17

2.B.3 Vacuum Failure

Micro-switches	Check with a qualified Electrician.
Solenoid Valve	Check if blocked or sticking and clean / replace accordingly.
Pipework	Clear / repair pipes and vacuum circuit. (See Section 4 Circuit Diagrams)
Pump	Check vacuum pump filters. (See Section 3.A.4 Maintenance)
Suction Cups	Replace as necessary
Insufficient Vacuum	Check gauge. Adjust regulating valve. Clean pump / replace cartridge.

2.B.4 Cartons Incorrectly Sealed (Glue System)

Insufficient / no glue	Check glue tank level. Check glue guns for incorrect setting. Check machine width is set correctly to carton height. Check glue is up to operating temperature. Check electrical signals from PEC's and solenoid valves are correct. Check machine width is set correctly to carton width. Check that glue line is in unvarnished surface of board.
Glue Line in Wrong Position	Check Glue Gun positions and adjust and adjust if necessary.
Carton Flaps Weak Sealed	Check that the adhesive has the correct open time. Check that the amount of glue is not too small or too light. Check that there is sufficient compression on the glue line. Check that the adhesive temperature is not set too low or too high.

2.B.4 Cartons Incorrectly Sealed (Tuck System)

Locking tab not forming on crease	Check tab folder / knife edge is set at crease line (Fig. 18)
Locking tab not locating	Check closing anvil is set correctly
	Check closing guide is set correctly

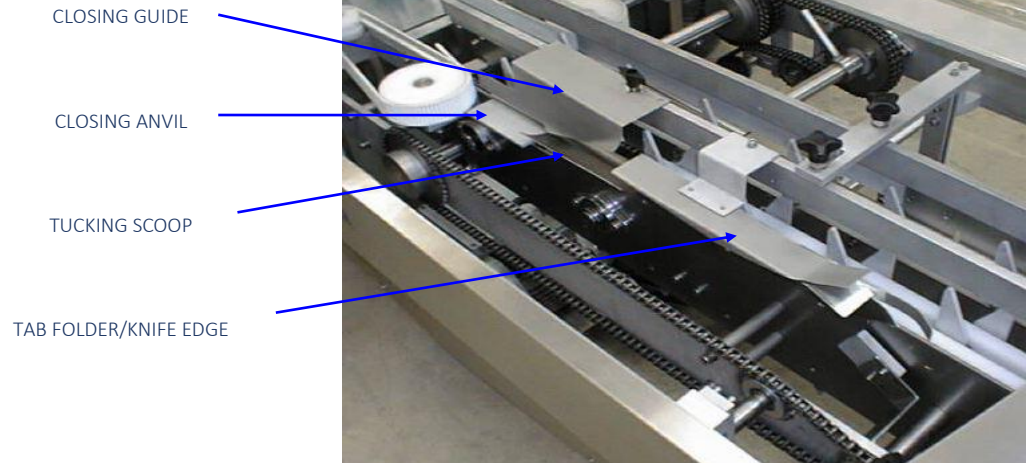


Fig. 18

2.B.5 Dust Flaps Incorrectly Closing

Rotary Tucker Timing out	Advance or retard the rotary tucker by loosening the two cap head screws and moving the tucker to the required position. The tucker should tuck in the trailing dust flaps and hold them in until the flap is under the guide. (Fig. 19)
--------------------------	---

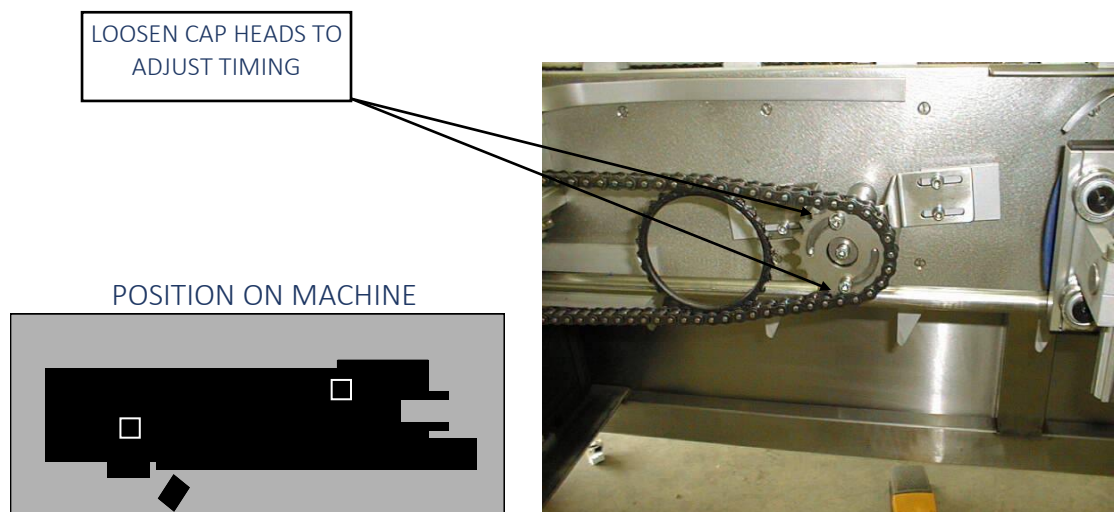


Fig. 19

2.B.6 Clutch Overload

Contamination on Chains	Check for deposits within the chain that have been passed on behind the chain runners. In this event, spray penetrating oil within the chain runners. Leave to soak for approximately 20-30 minutes. Reset the torque limiter and start the machine. (See section 2.B.1)
Product Jam	Check thoroughly throughout the machine and clear as necessary. Reset the torque limiter (Fig. 20) and start the machine. (See Section 2.B.1)
Carton Jam	Check thoroughly throughout the machine and clear as necessary. Reset the torque limiter and start the machine. (See Section 2.B.1)
Misalignment of settings	Check the setting and adjust as necessary.
Hint! Following a jam remove product or cartons which may be on the floor under the machine to prevent the machine picking up the discarded waste on restart.	

TORQUE LIMITER

LIMIT SWITCH

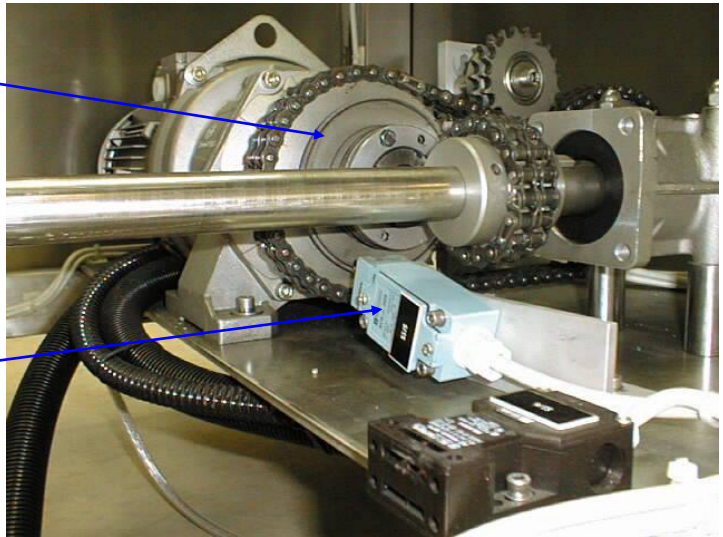


Fig. 20

SECTION 3

A MAINTENANCE SCHEDULE



WARNING!

SWITCH OFF ALL ELECTRICAL POWER BEFORE CARRYING OUT ANY ELECTRICAL MAINTENANCE



N.B. ALL ELECTRICAL SWITCHES AND RELAYS HAVE A LIFETIME OPERATION OF 500,000 OPERATIONS AND SHOULD BE CHECKED PERIODICALLY



WE RECOMMEND THAT OUR SERVICE ENGINEERS COMPLETE AN ANNUAL SERVICE. FOR FURTHER DETAILS PLEASE CONTACT OUR SERVICE DEPARTMENT

3.A.1 Lubrication Schedule

Main Drive Gearbox	Oil yearly with recommended gearbox oil.
Chains	Lightly oil as necessary.
Rack Shafts	Lightly oil as necessary.
Lead Screws	Lightly oil as necessary.
Pusher Arms / Yoke Plates	Lightly oil as necessary.

Note! Any parts of machinery that come into contact with the product should be lubricated with mineral based oils due to Product contamination; use 'Whitechain and Drive' or 'Food Lube' - spray when required.

3.A.2 Maintenance Schedule

Daily	Clean machine thoroughly by removing build up of dust, glue or product. It is essential that this schedule be maintained to ensure that the machine continues to run smoothly. (See Routine Cleaning Schedule Section 2.A.3) Check the water separator on the air pressure regulator unit and drain the unit accordingly. Check air pressure and maintain the compressors under their own schedule.
Weekly	Check machine thoroughly for efficient operation and security of components. Check vacuum suction cups for wear and tear so as to ensure accurate carton pick off. Check vacuum pipework for leaks.
Monthly	Product Infeed Tension, Remove the outer machine cladding and then tighten the tensioners as required. Ensure that the tensioners are given the same amount of tension on each side of the machine. Care should be taken not to over tighten the belt as this will induce excessive stretch (Fig. 21). Note! Ensure that any guarding that has been removed is replaced before the machine is put back into production.

POSITION ON MACHINE

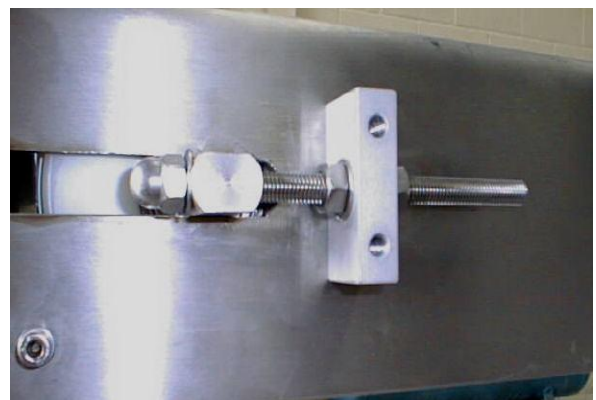
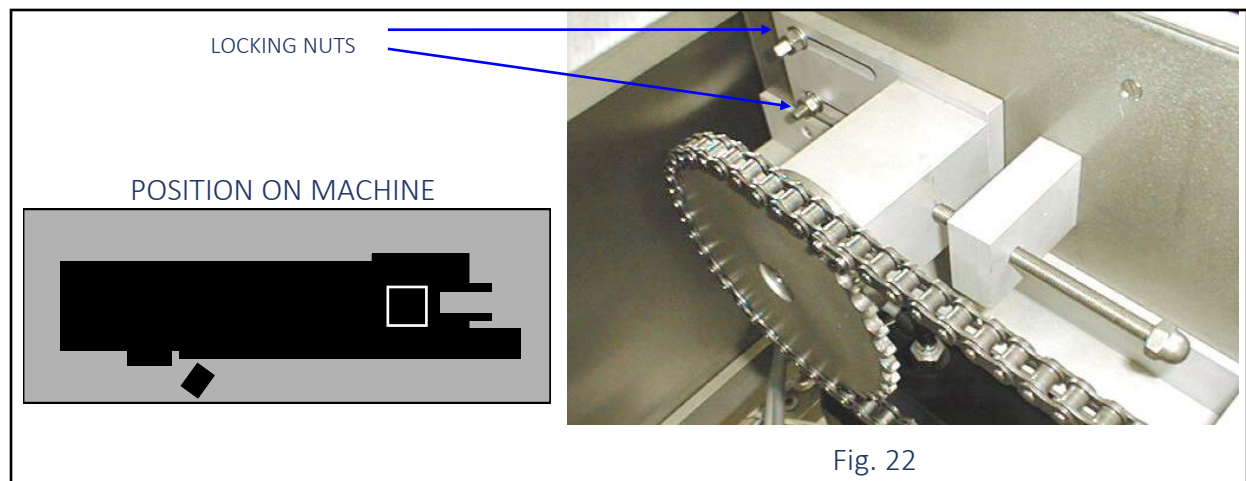


Fig. 21

Monthly Continued

Vacuum Pump	Clean and check the vacuum pump line filter if it is damaged replace with new.
Carton Flight Chain	Check flight conveyor chain tension and adjust as necessary by loosening the locking nuts and tightening in a clockwise direction. (Fig. 22) Check for wear and stretch on all chain links.



Machine Drives	Check all tensioners throughout the machine and tighten as required. Note! Care must be taken not to overtighten the tensioners, as this will induce excessive stretch on the chains. Ensure that any guarding that has been removed is replaced before the machine is put back into production.
Annually	Check all bearings for wear. All bearings are sealed for life and should require no lubrication.

3.A.3 Maintenance of the Air Pressure Regulator

CHECK	Pressure gauge to ensure that the required operating pressure is correct
CHECK	Pneumatic hoses, unions and connections are all secure and free from leaks

3.A.4 Maintenance of the Vacuum Pump

General Maintenance for Dry and Oil Lubricated Pumps	Filter Cartridges - Blow through from inside outwards. If found to be clogged with dust, grease or oil, filters are to be replaced. Use additional filters if dust collection is extremely high. Cooling Ducts - Clean with compressed air. Clogged ducts can cause overheating and the breakdown of the pump. Coupling Instructions - When the motor has to be replaced, it is essential that the correct coupling instructions be followed.
Maintenance Schedule By Pump Type Dry Pumps	Carbon Separator - Replace every 1000 - 2000 hours Vanes - Inspect the width of the vanes after 2000 hours for the first time and thereafter every 1000 hours. Clean the cylinder with dry compressed air. Install new vanes when the minimum width is reached. Bearings - Lubricate at given intervals with 4 to 5 shots of a grease gun, using Kluber Amblygon TA 15/2 whilst the pump is operational. The pump operates without oil; therefore, it must not be allowed to suck in water, oil or other liquids.
Oil Lubricated Pumps	Oil Level - When the oil has sunk half way down the glass, add fresh oil. Oil Change - First oil change after 100 hours, then thereafter every 500 - 1000 depending on the dust content of the intake air. Drain the oil whilst the pump is still warm by unscrewing the drain screw and refill up to the lip of the drain hole. Oil Separator Mesh - Clean the filter of oil separator in petroleum, then blow through with compressed air. Exhaust Air Filter - Check and replace as required.
Side Channel Blower	Suction Filter - Clean the suction filter every day or every week depending on dust content of the intake air. Clogged filters impair the pump and result in overload of the motor. Safety Valve - If safety valve is existing check weekly and clean as required. Cleaning - Clean compressor every month or more depending on the dust content of ambient air, to avoid overheating.
Clean Vacuum pump	Use manufacturers recommended safety solvents, under their instruction

3.A.5 Maintenance for the Electrical System



WARNING! SWITCH OFF ELECTRICAL POWER BEFORE CARRYING OUT ANY FORM OF MAINTENANCE



N.B. WE RECOMMEND THAT THE ELECTRICAL SYSTEM SHOULD BE CHECKED ON A REGULAR BASIS BY A QUALIFIED ELECTRICIAN

CHECK	That wiring and all electrically operated components are secure, undamaged and correctly enclosed Remake any damaged or defective wiring connections and replace any damaged components
CLEAN	Interior of control cabinet, taking care not to damage any item (Fig.23)
CHECK	Correct operation of safety guard interlocks

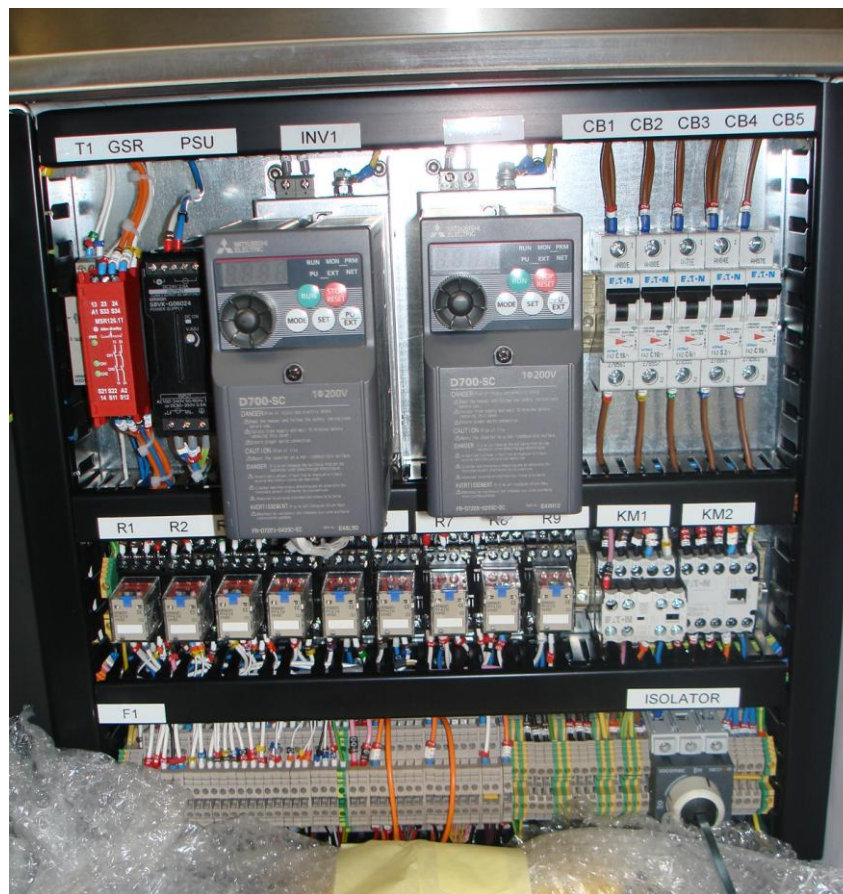


Fig. 23

B

SECTION 3

SIZE CHANGE

HINT!

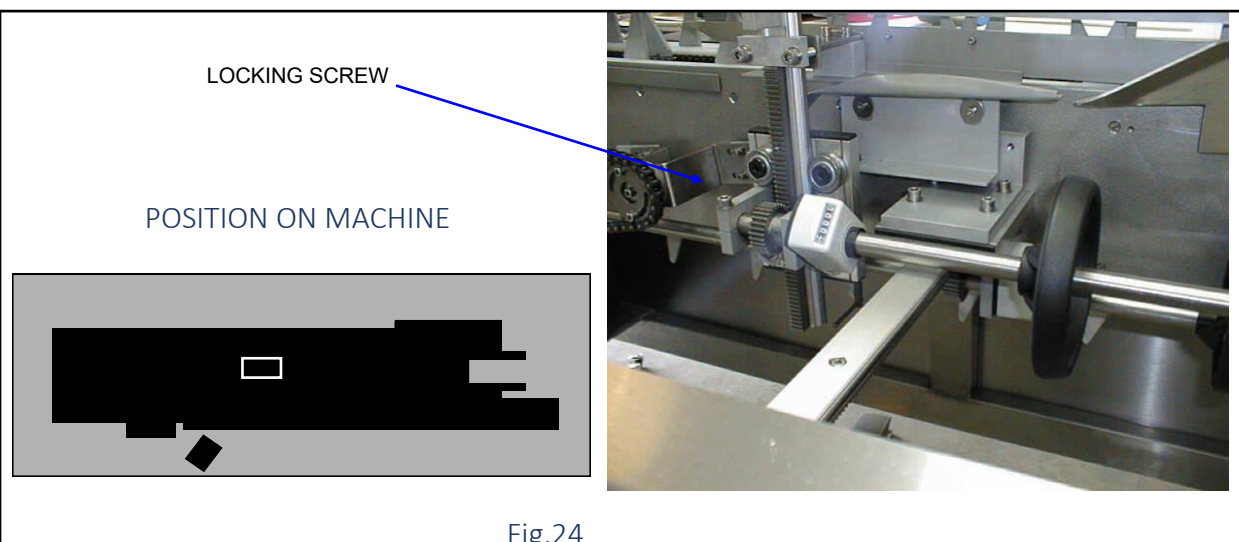
DURING THE CHANGE SIZE PROCEDURE OPERATORS WILL FIND IT ADVANTAGEOUS TO USE THE JOG FACILITY ON THE MACHINE.

3.B.1 Carton Change Procedures

For Height of Carton

Adjustment is obtained on the two overhead rails, by first loosening the locking screw. Then turning the hand wheel to the required readout on the digital indicator. (Fig. 24)

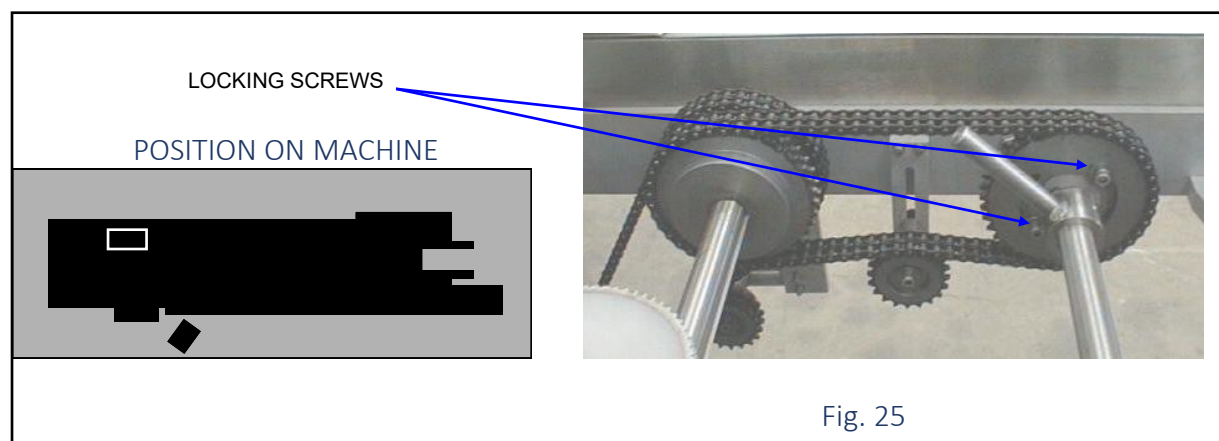
As a guideline the overhead rail should be positioned so as the carton can travel freely through the machine.



For Width of Carton (Flight Chains)

Adjustment is obtained by unlocking the clamping plate situated on the adjustable side of the machine on the end of the front flight drive shaft and then using the hand wheel turn the flight shaft and set to the required width of the carton.

As a guide the carton should be just held within the two sets of flights as illustrated below. (Fig. 25)



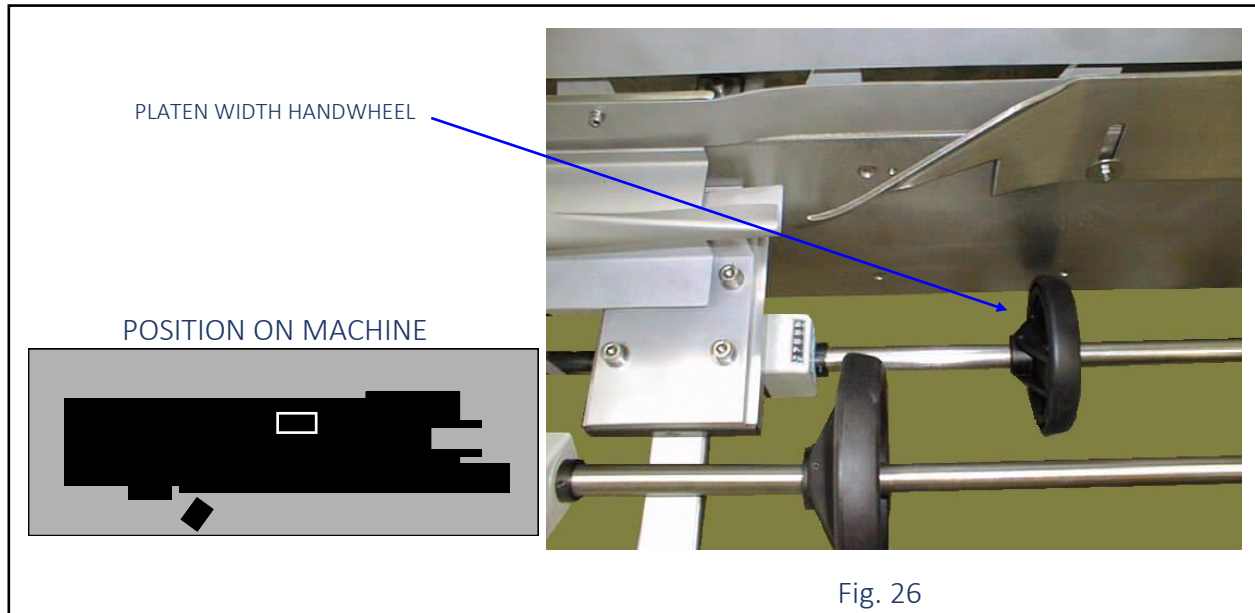
3.B.1 Cartoner Change Procedures Continued

For Between Flaps

Adjustment is obtained by winding the handle situated on the centre of the platen adjustment shaft to required size. Set with the aid of digital readout.

As a guide the carton should be just held by the side guides but be free to move through the machine.

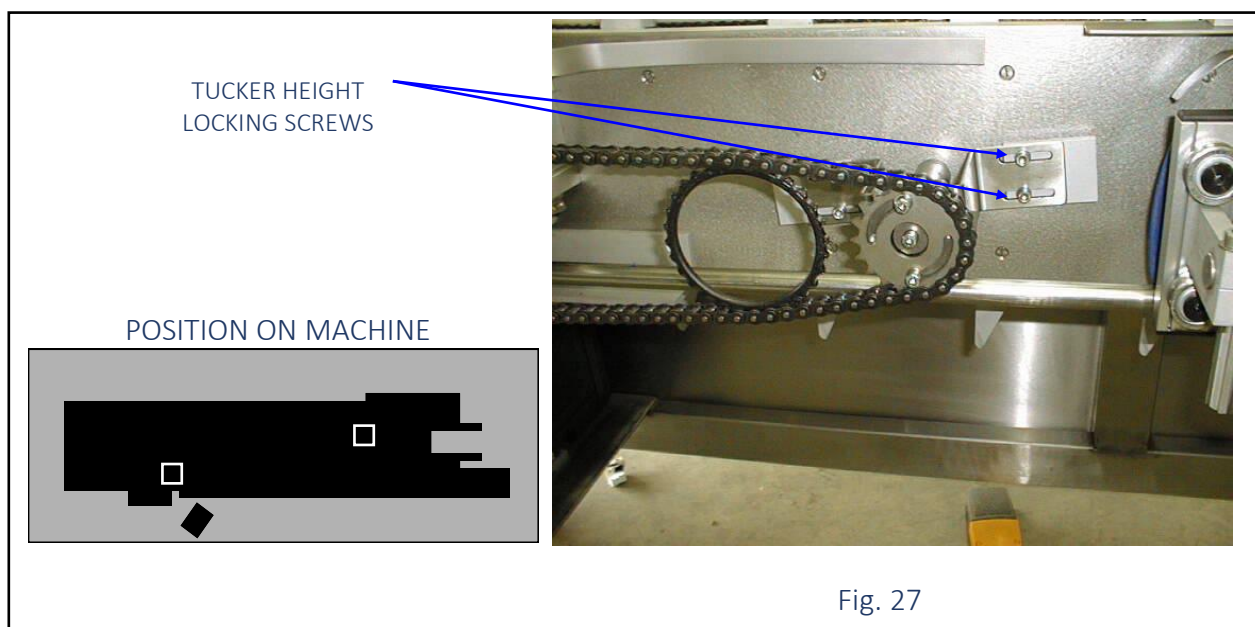
(Fig. 26)



For Dust Flaps

Under normal circumstances the rotary tucker will only need adjustment when the machine is changed from a very small carton to a very large carton.

If required, adjustment is obtained by raising or lowering rotary tucker. We recommend the use of the JOG facility during setting of the rotary tucker. (Fig. 27)



3.B.1 Cartoner Change Procedures Continued

For Flap Ploughs (Top and Bottom)

Adjustment is first obtained by undoing the locking screws and setting to the required position. Usually to within 10mm of the flap end. (Fig. 28)

POSITION ON MACHINE

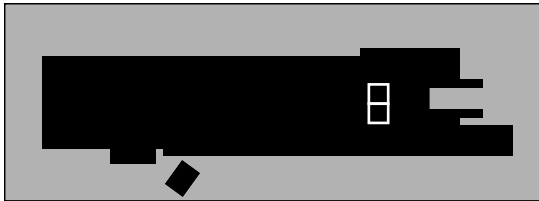


Fig. 28

Overhead Extensions

Adjustment is obtained by undoing the locking screws and setting to the required position by sliding the extension forward or back. (Fig. 29)

POSITION ON MACHINE



Fig. 29

3.B.2 Rotary Carton Feeder Change

Magazine Width

Adjustment for between flaps is made by undoing the brass locking screws situated on the magazine slides and adjusting to the required size. First set the carton crease to the edge of the fixed platen and adjust the other side to suit the full carton width. The magazine should be just loose enough to let the carton stack slide down through the magazine with ease. (Fig. 30)

BRASS LOCKING SCREW

POSITION ON MACHINE

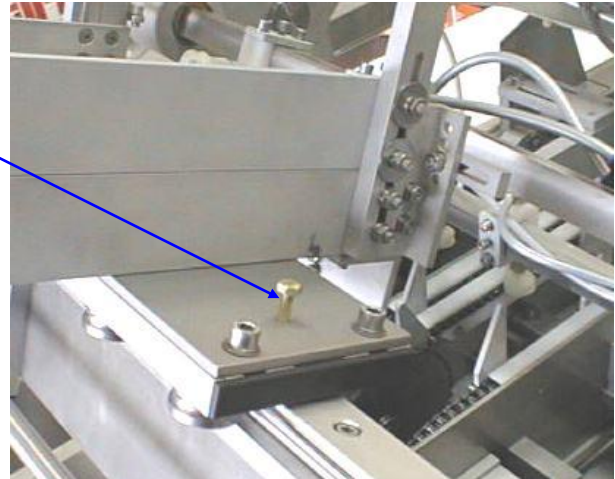


Fig. 30

Rotary Carton Height

Adjustment of the carton height is obtained by raising or lowering the rotary head, which will adjust the height of the suction cups to the carton. This is achieved by turning the hand wheel and winding the rotary up or down to give the required readout. To ensure correct carton placement onto the platen, make sure that when the arms are in the vertical position the flights are correctly positioned on the timing mark to receive carton. (Fig. 31/32)

POSITION ON MACHINE

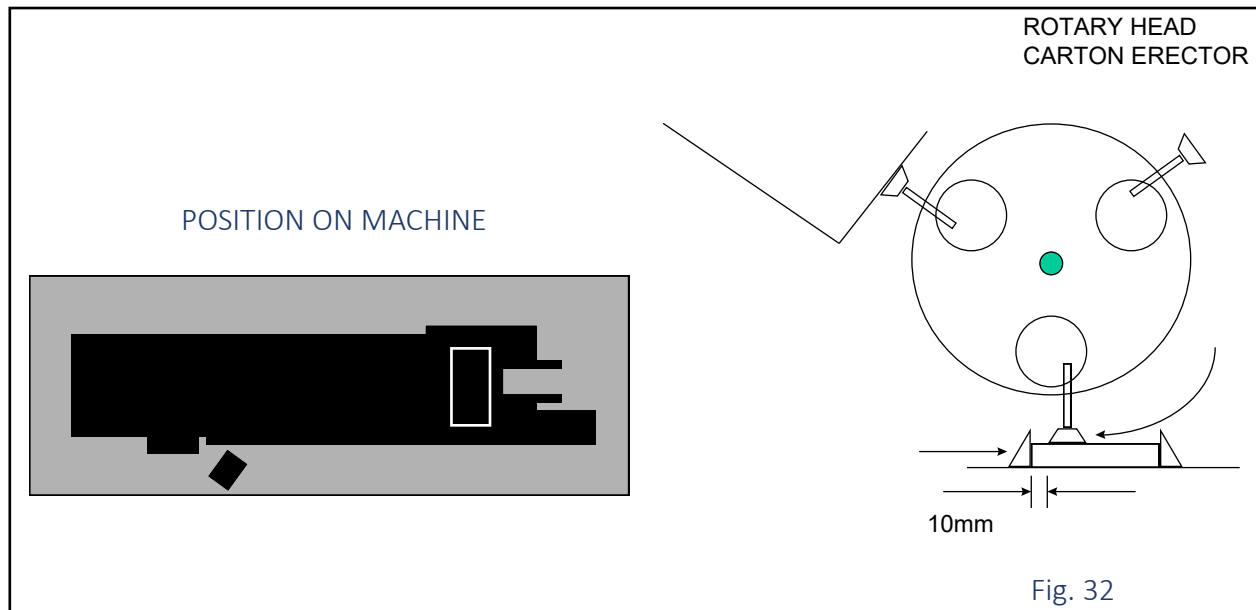


Fig. 31

3.B.2 Rotary Carton Feeder Change Continued

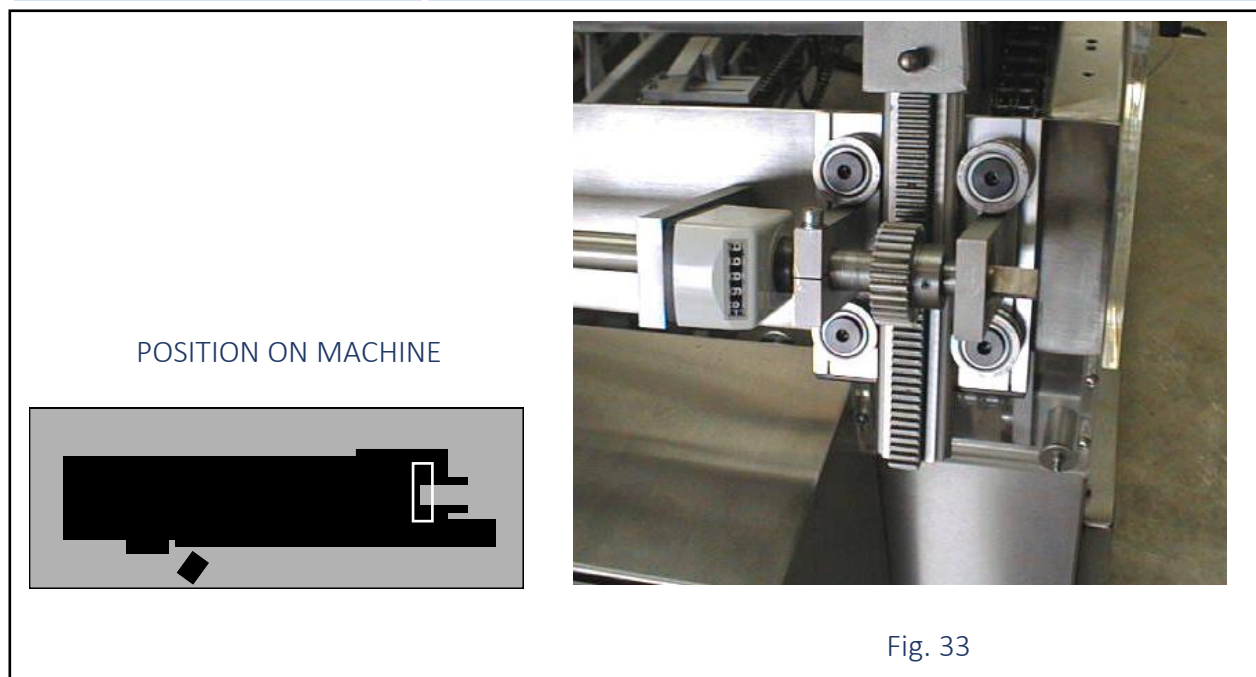
Rotary Carton Height continued

Subject to size and quality of cartons, when the suction cups are in the vertical position, the edge of the cup facing the main chain flight should be approximately 10mm away from the timing mark. (Fig. 32)



Magazine Height

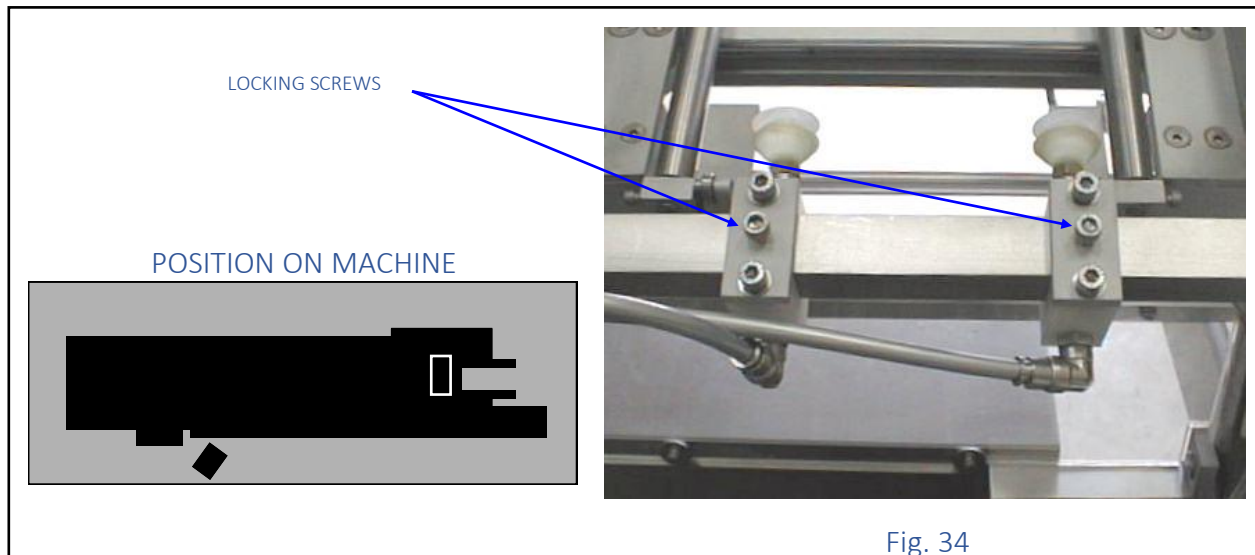
Firstly, by using the 'Jog' button, position suction cups so that they 'squarely' touch the cartons in the magazine. By using a spanner on the adjuster situated beneath the magazine turn to achieve the required read-out on the digital counter (Fig. 33)



3.B.2 Rotary Carton Feeder Change Continued

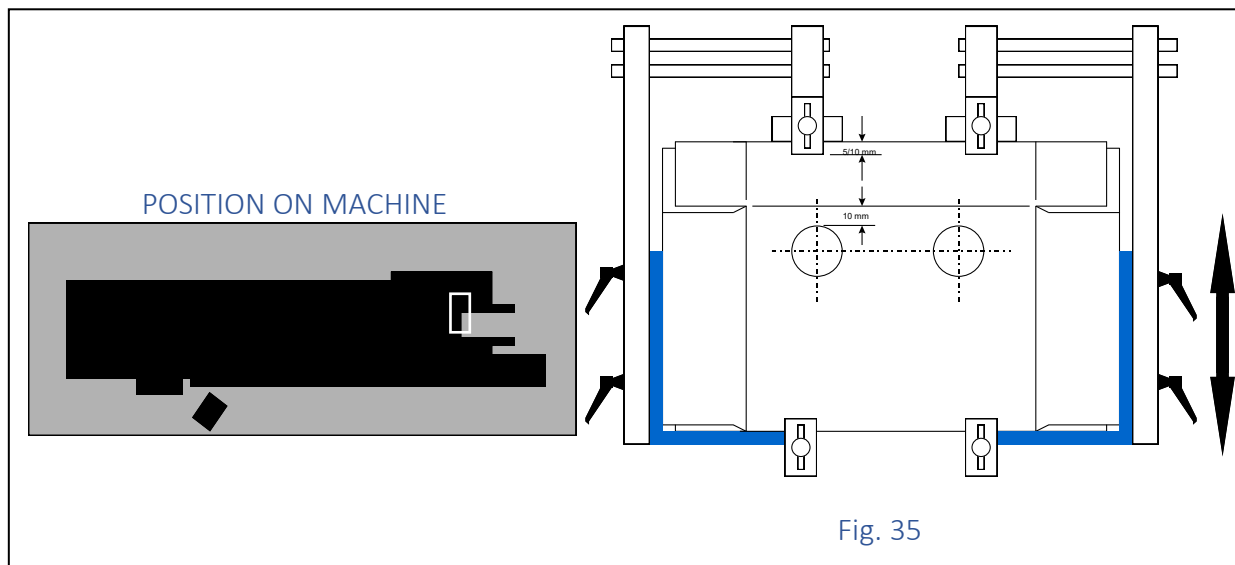
Magazine Height Continued

Adjustment of the suction cups should now be made to suit the width of the carton. This is achieved by undoing the locking screw situated on the suction cup holder and moving to the correct position. Ensure that suction cups miss the main chain flights during the operation by using the 'Jog' button to position the suction cups in between the flights. (Fig. 34)



Magazine Height Continued

Finally, set the top Carton Rail to the required height by undoing the locking handles and positioning accordingly and if required adjust the ledges to suit, approximately 5-10mm. (See Fig. 35)



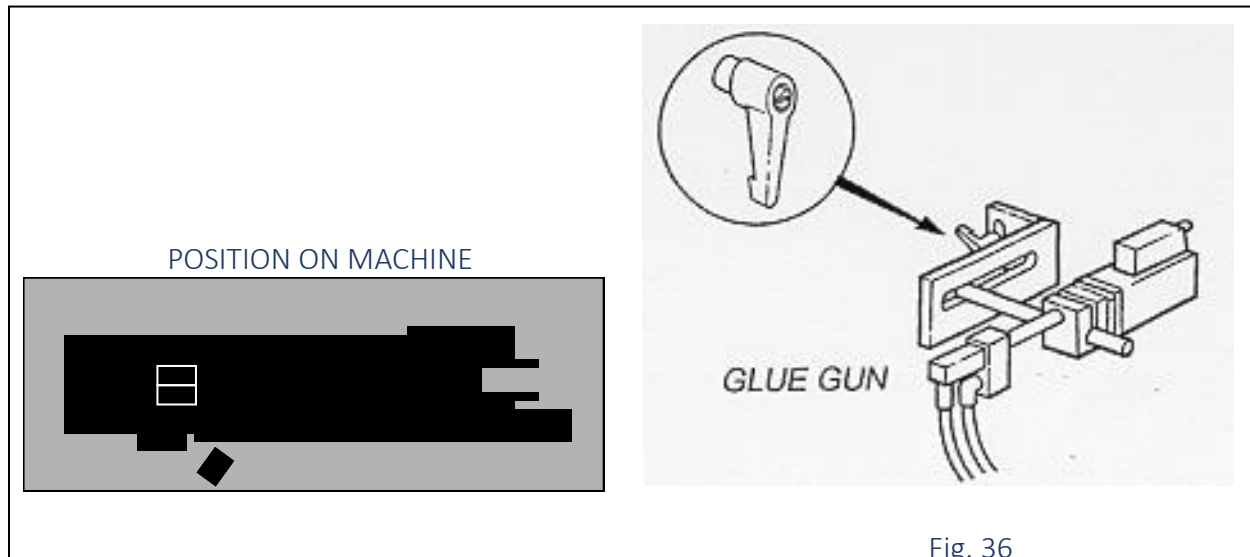
3.B.3 Carton Gluing Change

PLEASE REFER TO NORDSON HOT MELT APPLICATOR MANUAL

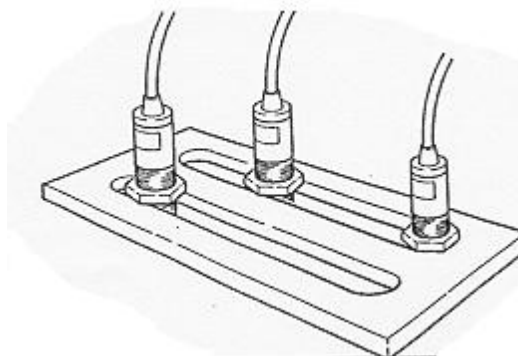
Glue Unit Position

Adjustment of the glue gun position is achieved by loosening the two locking handles and moving the gun up or down to the required position

As a guide the glue guns should be applying glue to the minor flap so as when the major flap is ploughed down the glue line is approx. 6mm from the edge. (Fig. 36)



The P.E.C.'s that control the glue line length and position are factory set and should require no adjustment. If adjustment is required loosen the locking nuts and slide the P.E.C.'s to the new position. (Fig. 37)



WARNING !

THE GLUE UNIT WILL BE HOT DO NOT TOUCH WITHOUT ADEQUATE PROTECTIVE CLOTHING

3.B.3 Carton Tucking Change

Tab folder / Knife edge

These items are either changed out or adjusted depending upon machine specification. Adjustable parts should be set to the carton height (Fig. 41). All other parts are factory set and require no adjustment.

CLOSING GUIDE
CLOSING ANVIL
TUCKING SCOOP
TAB FOLDER/KNIFE EDGE

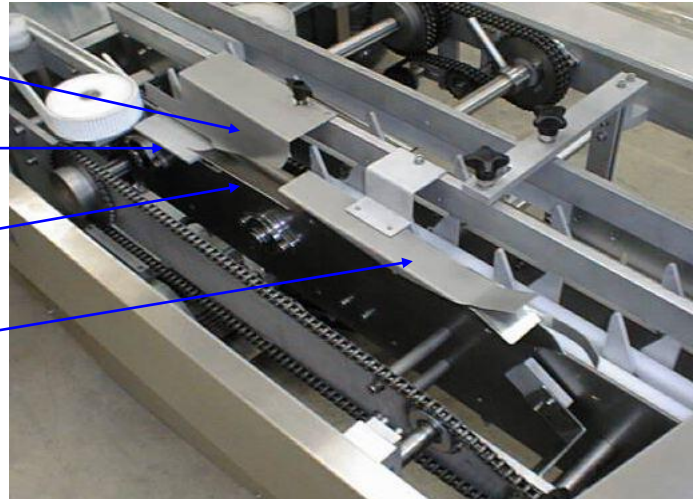


Fig.40

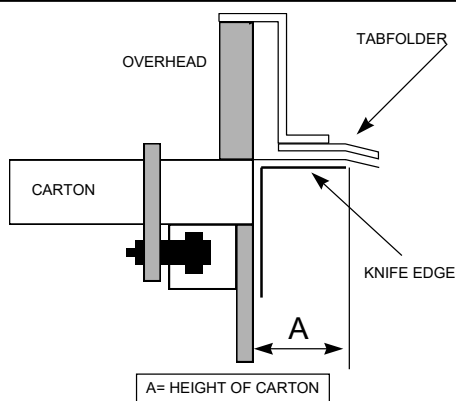


Fig. 41

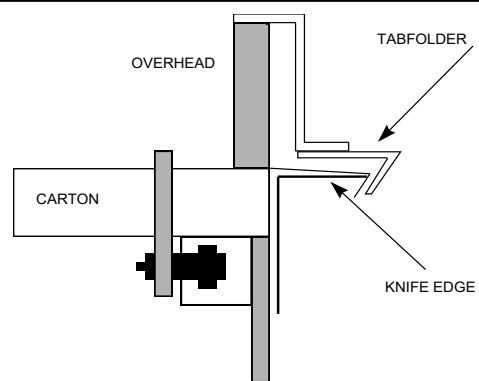


Fig.42

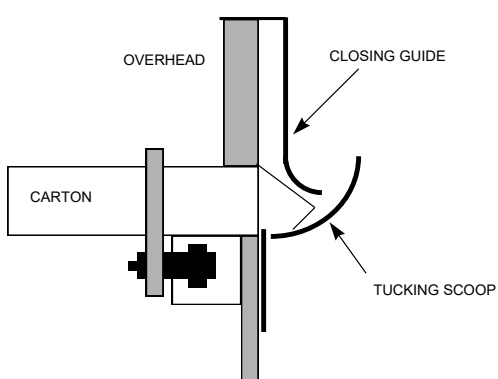


Fig.43

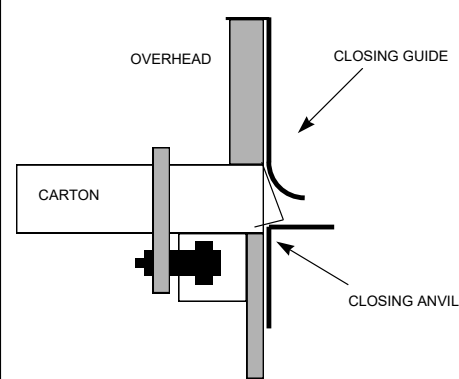


Fig. 44

SECTION 4

A HOT MELT ADHESIVE

Introduction

In their solid state these products are generally free of hazard. However, in their molten state there are potential hazards whose significance depends on the individual situation, particularly since the amount of product involved varies widely from one application to another.

Hazards

Burns

Burns are undoubtedly the major source of injury and they could be severe since application temperatures are usually in the range 120°C-240°C. They may result from either contact with the product itself or from contact with equipment used for handling or applying it.

Fumes

Under normal operating conditions in well-ventilated areas, fumes do not constitute a serious hazard. They are, however, composed of low molecular weight organic materials which are unpleasant and which may be harmful or irritating to the respiratory system. If the vapour concentration is high enough, they may constitute an explosion or fire hazard.

Hot melt products degrade to varying extents from the effect of heat and oxygen and the effects are directly related to time, temperature and formulation. When grossly overheated (at least 30°C-40°C above the recommended operating temperature), the effects can be severe and some of the decomposition products could pose a serious risk to health.

Fire

There is no particular fire hazard with hot melts, but in a fire situation they are combustible and will burn with the evolution of black acrid smoke. It should be noted that a degraded hot melt generally has a lower auto-ignition temperature than the fresh product.

Essential Safety Precautions

Burns

Avoid splashing and take special care when stirring or adding product to a reservoir. Do not allow molten product to come into contact with the skin or eyes. Face and eye protectors conforming to BS.2092 must be worn when there is any possibility of splashing occurring.

Protective clothing must be kept in good condition and worn at all times when handling molten adhesive or hot equipment.

Fumes

DO NOT breath fumes.

Adequate extraction must be provided to ensure exposure is kept as low as is reasonably practicable. See HSE Guidance Notes EH18, EH22 and EH40 (revised annually).

Cleanliness: the equipment should be kept clean and free from build-up of solidified material, which on degradation could possibly form harmful or irritating vapours.

Fire

Ventilation ducts must be inspected regularly and kept clean to prevent the build-up of condensate, which could produce a fire or explosion risk.

Storage

Every precaution must be taken to avoid hot melt adhesives or sealant becoming moist from condensation or other sources since adding wet material to already molten products is likely to cause spitting.

Emergency Actions

Burns

Immediately treat the affected area by immersion or irrigation with clean cold water until the adhesive or sealant is cold.

DO NOT ATTEMPT TO REMOVE THE ADHESIVE OR SEALANT FROM THE SKIN.

Cover the affected area with a cold compress or dry sterile gauze dressing.

Obtain medical attention immediately.

Fumes

If exposure does result in discomfort, the affected person should be removed to fresh air, kept warm and at rest. Medical advice should be sought.

Fires

Switch off source of heat.

Small fires may be tackled with dry powder or carbon dioxide extinguishers.

DO NOT USE WATER.